HDOC Day 34

Aim

To write a menu-driven C program using separate functions and variable-length 2D arrays to verify the left and right distributive properties of matrix multiplication over addition, display both resulting matrices, and report whether each property is verified.

Theory

Matrix addition is performed element by element for matrices of equal order. Matrix multiplication of an (m x n) matrix and an (n x p) matrix produces an (m x p) matrix. The left distributive law is A(B+C) = AB+AC, where A is m x n and B,C are n x p. The right distributive law is (A+B)C = AC+BC, where A,B are m x n and C is n x p.

Algorithm

1. Start and display the menu: left distributive, right distributive, exit.

2. Read the choice. For choices 1 or 2, input positive dimensions m, n and p.

3. For left distributive property, read A(m x n), B(n x p), C(n x p). Compute B+C; multiply A by this sum to get LHS. Separately compute AB and AC and add them to obtain RHS.

4. For right distributive property, read A(m x n), B(m x n), C(n x p). Compute A+B; multiply this sum by C to get LHS. Separately compute AC and BC and add them to obtain RHS.

5. Display the LHS and RHS matrices of order m x p; compare corresponding elements and print whether the property is verified.

6. Return to the menu until Exit is selected; stop.

C Program

```
#include <stdio.h>

void read(int r,int c,int a[r][c]) {
    for(int i=0;i<r;i++)
        for(int j=0;j<c;j++) scanf("%d",&a[i][j]);
}
void add(int r,int c,int a[r][c],int b[r][c],int d[r][c]) {
    for(int i=0;i<r;i++)
        for(int j=0;j<c;j++) d[i][j]=a[i][j]+b[i][j];
}
void multiply(int r,int n,int c,int a[r][n],
              int b[n][c],int d[r][c]) {
    for(int i=0;i<r;i++)
        for(int j=0;j<c;j++) {
            d[i][j]=0;
            for(int k=0;k<n;k++) d[i][j]+=a[i][k]*b[k][j];
        }
}
void show(int r,int c,int a[r][c]) {
    for(int i=0;i<r;i++) {
        for(int j=0;j<c;j++) printf("%d ",a[i][j]);
        printf("\n");
    }
}
void verify(int r,int c,int a[r][c],int b[r][c]) {
    int ok=1;
    printf("LHS:\n"); show(r,c,a);
    printf("RHS:\n"); show(r,c,b);
    for(int i=0;i<r;i++)
```

```
        for(int j=0;j<c;j++)
            if(a[i][j]!=b[i][j]) ok=0;
    printf("Property %sverified.\n",ok?"":"not ");
}
void left(int m,int n,int p) {
    int A[m][n],B[n][p],C[n][p],S[n][p];
    int L[m][p],R[m][p],X[m][p],Y[m][p];
    printf("Enter A (%d x %d):\n",m,n); read(m,n,A);
    printf("Enter B (%d x %d):\n",n,p); read(n,p,B);
    printf("Enter C (%d x %d):\n",n,p); read(n,p,C);
    add(n,p,B,C,S); multiply(m,n,p,A,S,L);
    multiply(m,n,p,A,B,X); multiply(m,n,p,A,C,Y);
    add(m,p,X,Y,R); verify(m,p,L,R);
}
void right(int m,int n,int p) {
    int A[m][n],B[m][n],C[n][p],S[m][n];
    int L[m][p],R[m][p],X[m][p],Y[m][p];
    printf("Enter A (%d x %d):\n",m,n); read(m,n,A);
    printf("Enter B (%d x %d):\n",m,n); read(m,n,B);
    printf("Enter C (%d x %d):\n",n,p); read(n,p,C);
    add(m,n,A,B,S); multiply(m,n,p,S,C,L);
    multiply(m,n,p,A,C,X); multiply(m,n,p,B,C,Y);
    add(m,p,X,Y,R); verify(m,p,L,R);
}
int main(void) {
    int choice,m,n,p;
    do {
        printf("\n1. Left distributive\n");
        printf("2. Right distributive\n3. Exit\n");
        printf("Enter choice: ");
        if(scanf("%d",&choice)!=1) return 1;
        if(choice==1 || choice==2) {
            printf("Enter m n p: ");
            if(scanf("%d%d%d",&m,&n,&p)!=3) return 1;
            if(m<=0 || n<=0 || p<=0) {
                printf("Invalid dimensions.\n"); continue;
            }
            if(choice==1) left(m,n,p);
            else right(m,n,p);
        } else if(choice!=3) printf("Invalid choice.\n");
    } while(choice!=3);
    return 0;
}
```

Sample Output

```
1. Left distributive
2. Right distributive
3. Exit
Enter choice: 1
Enter m n p: 2 2 2
Enter A (2 x 2):
1 2
3 4
Enter B (2 x 2):
1 0
0 1
Enter C (2 x 2):
2 1
1 2
LHS:
5 5
13 11
RHS:
5 5
13 11
Property verified.

1. Left distributive
2. Right distributive
```

```
3. Exit
Enter choice: 2
Enter m n p: 2 2 2
Enter A (2 x 2):
1 2
3 4
Enter B (2 x 2):
1 0
0 1
Enter C (2 x 2):
2 1
1 2
LHS:
6 9
12 17
RHS:
6 9
12 17
Property verified.

Enter choice: 3
```

Result

The menu-driven C program was executed successfully. Both A(B+C) = AB+AC and (A+B)C = AC+BC were verified by comparing and displaying the left-hand-side and right-hand-side matrices.